SECTION-I: MATHEMATICAL ABILITY — Solutions & Explanations

Infosys Systems Engineer (SE) | 5-6 Nov 2026 Drive | 100% Pure SE Verified

Total Questions: 11 | **Total Marks:** 11

Section: Technical ability/Quantitative Aptitude

Problem 1 • A code word is to consist of two distinct English alphabets followed by two distinct numbers from1 and 9 (both Inclusive). For example, CA23 is a code word. How many such code words are there?

PRACTICE VARIANT

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard

A code word is to consist of two distinct English alphabets followed by two distinct numbers from1 and 9 (both Inclusive). For example, CA23 is a code word. How many such code words are there?

- A 615800
- C 719500
- D 410800

Verified Answer: Option (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 2 • In an entrance exam of 200 objective questions, a student can score 1 point for every correct answer, loss 1/4 points for every wrong answer and loss 1/2 point for every unanswered question. If he attempts only 160 questions and he scores 100 points then the number of questions answered by him correctly is:

CORE SE

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

In an entrance exam of 200 objective questions, a student can score 1 point for every correct answer, loss 1/4 points for every wrong answer and loss 1/2 point for every unanswered question. If he attempts only 160 questions and he scores 100 points then the number of questions answered by him correctly is:

- A A. 132
- B A. 126
- C A. 139

Verified Answer: Option (D) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (D)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 3 • Two containers of milk contain mixtures of water and milk in ratio 5:4 and 7:9. In what ratio they should be mixed so that mixture is of 6:6 ratios?

○ PRACTICE VARIANT

ID: Q3 | Source: Infosys PYQ | Level: Infosys SE Standard

Two containers of milk contain mixtures of water and milk in ratio 5:4 and 7:9. In what ratio they should be mixed so that mixture is of 6:6 ratios?

A A. 6:5

B A. 7:5

C A. 8:9

✓ **Verified Answer:** Option (D) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (D)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 4 • There are two pipes in a tank. Pipe A is for filling the tank and pipe b is for emptying the tank. If A can fill the tank in 10 hours and b can empty the tank in 15 hours then find how many hours will it take to completely fill a half empty tank?

○ PRACTICE VARIANT

ID: Q4 | Source: Infosys PYQ | Level: Infosys SE Standard

There are two pipes in a tank. Pipe A is for filling the tank and pipe b is for emptying the tank. If A can fill the tank in 10 hours and b can empty the tank in 15 hours then find how many hours will it take to completely fill a half empty tank?

A A. 30 hours

C A. 20 hours

D A. 33.33 hours

✓ **Verified Answer:** Option (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 5 • A good costs Rs 12000/-. A customer can choose either of the pairs of successive discounts. Pair A consists of successive discounts of 20% & 30% and pair B consists of 40% & 10%. Customer X chose A & customer Y choose B. What is the difference between their payments

○ PRACTICE VARIANT

ID: Q5 | Source: Infosys PYQ | Level: Infosys SE Standard

A good costs Rs 12000/-. A customer can choose either of the pairs of successive discounts. Pair A consists of successive discounts of 20% & 30% and pair B consists of 40% & 10%. Customer X chose A & customer Y choose B. What is the difference between their payments

A A) Rs 120

C A) Rs 180

D A) Rs 60

✓ **Verified Answer:** Option (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 6 • A towel was 40 cm broad and 90 cm long. When bleached, it was found to have lost 20% of its length and 10% of its width. The percentage decrease in area is:

○ PRACTICE VARIANT

ID: Q6 | Source: Infosys PYQ | Level: Infosys SE Standard

A towel was 40 cm broad and 90 cm long. When bleached, it was found to have lost 20% of its length and 10% of its width. The percentage decrease in area is:

B 30%

C 32%

D 33%

✓ **Verified Answer:** Option (A) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (A)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 7 • The total time taken by a boat to cover 'D+25' km downstream and same distance in upstream is 20 hours. If speed of boat is 15 km/hr. in still water and speed of stream is 40% less than speed of boat, then find the value of D

PRACTICE VARIANT

ID: Q7 Source: Infosys PYQ Level: Infosys SE Standard

The total time taken by a boat to cover 'D+25' km downstream and same distance in upstream is 20 hours. If speed of boat is 15 km/hr. in still water and speed of stream is 40% less than speed of boat, then find the value of D

A 66 km

B 78 km

C 62km

Verified Answer: Option (D) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (D)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 8 • Direction : There are seven friends - Brian, Paul, Qasim, Rustam, Sam, Tom and Winston - each of whom likes a different subject among Chemistry, Geography, Physics, Math Sociology, History and Biology. They are ranked 1 to 7 in their final exams, with 1 being the best rank and 7 the worst. With the help of the information given below you have to deduce their respective ranks and the subjects they like. Qasim likes Sociology, and he stands above Tom but below Winston. Tom gets the worst rank, and he likes neither Geography nor History. Sam likes Physics, and he stands above both Qasim and Winston but below Paul Rustam gets Rank 2, and he likes neither Chemistry nor Math. Paul likes Biology, and he hasn't secured the best rank. The student who likes maths gets the best rank. Rustam doesn't like Geography Question8 : Which of the following pairs is correct for Rustam?

PRACTICE VARIANT

ID: Q8 Source: Infosys PYQ Level: Infosys SE Standard

Direction : There are seven friends - Brian, Paul, Qasim, Rustam, Sam, Tom and Winston - each of whom likes a different subject among Chemistry, Geography, Physics, Math Sociology, History and Biology. They are ranked 1 to 7 in their final exams, with 1 being the best rank and 7 the worst. With the help of the information given below you have to deduce their respective ranks and the subjects they like. Qasim likes Sociology, and he stands above Tom but below Winston. Tom gets the worst rank, and he likes neither Geography nor History. Sam likes Physics, and he stands above both Qasim and Winston but below Paul Rustam gets Rank 2, and he likes neither Chemistry nor Math. Paul likes Biology, and he hasn't secured the best rank. The student who likes maths gets the best rank. Rustam doesn't like Geography Question8 : Which of the following pairs is correct for Rustam?

B Rank 3 – Chemistry

C Rank 4 – Physics

D Rank 5 – Biology

Verified Answer: Option (A) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (A)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 9 • Direction : There are seven friends - Brian, Paul, Qasim, Rustam, Sam, Tom and Winston - each of whom likes a different subject among Chemistry, Geography, Physics, Math Sociology, History and Biology. They are ranked 1 to 7 in their final exams, with 1 being the best rank and 7 the worst. With the help of the information given below you have to deduce their respective ranks and the subjects they like. Qasim likes Sociology, and he stands above Tom but below Winston. Tom gets the worst rank, and he likes neither Geography nor History. Sam likes Physics, and he stands above both Qasim and Winston but below Paul Rustam gets Rank 2, and he likes neither Chemistry nor Math. Paul likes Biology, and he hasn't secured the best rank. The student who likes maths gets the best rank. Rustam doesn't like Geography Question 9 : Which subject is liked by the student who got exactly the middle rank (Rank 4)?

PRACTICE VARIANT

ID: Q9 Source: Infosys PYQ Level: Infosys SE Standard

Direction : There are seven friends - Brian, Paul, Qasim, Rustam, Sam, Tom and Winston - each of whom likes a different subject among Chemistry, Geography, Physics, Math Sociology, History and Biology. They are ranked 1 to 7 in their final exams, with 1 being the best rank and 7 the worst. With the help of the information given below you have to deduce their respective ranks and the subjects they like. Qasim likes Sociology, and he stands above Tom but below Winston. Tom gets the worst rank, and he likes neither Geography nor History. Sam likes Physics, and he stands above both Qasim and Winston but below Paul Rustam gets Rank 2, and he likes neither Chemistry nor Math. Paul likes Biology, and he hasn't secured the best rank. The student who likes maths gets the best rank. Rustam doesn't like Geography Question 9 : Which subject is liked by the student who got exactly the middle rank (Rank 4)?

B History

C Chemistry

D Sociology

✓ **Verified Answer:** Option (A) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (A)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 10 • Direction : There are seven friends - Brian, Paul, Qasim, Rustam, Sam, Tom and Winston - each of whom likes a different subject among Chemistry, Geography, Physics, Math Sociology, History and Biology. They are ranked 1 to 7 in their final exams, with 1 being the best rank and 7 the worst. With the help of the information given below you have to deduce their respective ranks and the subjects they like. Qasim likes Sociology, and he stands above Tom but below Winston. Tom gets the worst rank, and he likes neither Geography nor History. Sam likes Physics, and he stands above both Qasim and Winston but below Paul Rustam gets Rank 2, and he likes neither Chemistry nor Math. Paul likes Biology, and he hasn't secured the best rank. The student who likes maths gets the best rank. Rustam doesn't like Geography Question 10: Which of the following is the correct sequence of students in consecutive ranks (from higher to lower)?

○ PRACTICE VARIANT

ID: Q10 Source: Infosys PYQ Level: Infosys SE Standard

Direction : There are seven friends - Brian, Paul, Qasim, Rustam, Sam, Tom and Winston - each of whom likes a different subject among Chemistry, Geography, Physics, Math Sociology, History and Biology. They are ranked 1 to 7 in their final exams, with 1 being the best rank and 7 the worst. With the help of the information given below you have to deduce their respective ranks and the subjects they like. Qasim likes Sociology, and he stands above Tom but below Winston. Tom gets the worst rank, and he likes neither Geography nor History. Sam likes Physics, and he stands above both Qasim and Winston but below Paul Rustam gets Rank 2, and he likes neither Chemistry nor Math. Paul likes Biology, and he hasn't secured the best rank. The student who likes maths gets the best rank. Rustam doesn't like Geography Question 10: Which of the following is the correct sequence of students in consecutive ranks (from higher to lower)?

A Winston – Paul – Sam – Qasim – Tom

B Sam – Paul – Winston – Qasim – Tom

D Paul – Winston – Sam – Qasim – Tom

✓ **Verified Answer:** Option (C) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (C)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 11 • To solve a crime, police interrogate both Felix and Victor. Felix speaks the truth in 60% of cases, and Victor in 40% of cases. Police can catch the criminal only if at least one of them speaks the truth. What is the probability that the police catch the criminal?

○ PRACTICE VARIANT

ID: Q11 Source: Infosys PYQ Level: Infosys SE Standard

To solve a crime, police interrogate both Felix and Victor. Felix speaks the truth in 60% of cases, and Victor in 40% of cases. Police can catch the criminal only if at least one of them speaks the truth. What is the probability that the police catch the criminal?

A 13/25

B 21/25

C 6/25

✓ **Verified Answer:** Option (D) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (D)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.